Day 3: Operating Systems

1. Definition and Importance

- An **Operating System (OS)** is the foundation of all computing.
- It acts as a **resource manager** (CPU, memory, storage, devices) and a **service provider** (user interface, security, networking).
- Without an OS, hardware cannot interact with users or applications.

2. Core Functions of an OS (Expanded)

- **Process Management**
  - OS creates a process when you open an application.
  - It schedules CPU time using algorithms (e.g., Round Robin, Priority Scheduling).
  - Prevents deadlocks by controlling resource allocation.
- **Memory Management**
  - Allocates RAM to active programs.
  - Uses **virtual memory** (hard disk space) when RAM is insufficient.
  - Frees memory when programs close.
- **File System Management**
  - Organizes files into hierarchical structures (folders, subfolders).
  - Enforces permissions (read, write, execute).
  - Examples: NTFS (Windows), HFS+ (macOS), ext4 (Linux).
- **Device Management**
  - Uses **drivers** to communicate with hardware.
  - Manages input/output queues.
- **Security**
  - Provides authentication (passwords, biometrics).
  - Authorization (admin vs standard users).
  - Encryption for sensitive data.
- **Networking**
  - Provides TCP/IP stack for internet communication.
  - Manages Wi-Fi/Ethernet connections.
- **User Interface**
  - GUI: Windows/macOS (icons, menus, windows).

    - CLI: Linux terminal, PowerShell.

3. Types of Operating Systems

- **Batch OS**: Executes jobs without user interaction.
- **Time-sharing OS**: Multiple users share resources simultaneously.
- **Distributed OS**: Manages multiple computers as one system.
- **Embedded OS**: Runs on devices like ATMs, cars, medical equipment.
- **Mobile OS**: Android, iOS — optimized for touch and mobility.

4. Examples of OS in Practice

- **Windows**: Dominates business and personal computing.
- **macOS**: Preferred in creative industries.
- **Linux**: Powers servers, cybersecurity tools, supercomputers.
- **Android/iOS**: Run billions of smartphones worldwide.

5. Practical Exercises (Day 3)

- **Navigating Windows Desktop**
    - Step 1: Log in.
    - Step 2: Identify Start menu, taskbar, icons.
    - Step 3: Open two applications and switch between them.
- **Accessing System Settings**
    - Step 1: Press Win + I.
    - Step 2: Explore categories (System, Devices, Network).
    - Step 3: Change desktop background.
- **Checking System Information**
    - Step 1: Right-click “This PC” → Properties.
    - Step 2: Note processor type, RAM size, OS version.
    - Step 3: Compare with another computer.

⚙️ Day 4: Device Operations

1. Proper Startup and Shutdown Procedures

- **Startup**
    - Step 1: Press power button.
    - Step 2: BIOS/UEFI runs POST (hardware check).
    - Step 3: Bootloader loads OS kernel.

    - Step 4: OS initializes drivers and services.
    - Step 5: Login screen appears → enter credentials.
- **Shutdown**
    - Step 1: Save all work.
    - Step 2: Click Start → Power → Shut down.
    - Step 3: Wait until system powers off completely.

⚠ Improper shutdown (holding power button) risks data loss and hardware damage.

2. Sleep, Hibernate, Restart

- **Sleep**: Saves state in RAM, quick resume.
- **Hibernate**: Saves state to disk, no power used, slower resume.
- **Restart**: Reloads OS kernel, clears temporary memory.

Steps (where possible)

- Sleep: Start → Power → Sleep → Resume with key press.
- Hibernate: Start → Power → Hibernate → Resume with power button.
- Restart: Start → Power → Restart → System reloads OS.

3. User Accounts and Login Procedures

- **Creating a New Account**
    - Step 1: Open Settings → Accounts → Family & other users.
    - Step 2: Click "Add account."
    - Step 3: Choose Administrator or Standard User.
    - Step 4: Set password or PIN.
    - Step 5: Log out and log in with new account.
- **Login Methods**: Password, PIN, biometrics, two-factor authentication.

4. Basic Troubleshooting Concepts

- **Frozen Application**
    - Step 1: Press Ctrl + Shift + Esc.
    - Step 2: Open Task Manager.
    - Step 3: Select frozen app → End Task.
- **Unresponsive System**
    - Step 1: Press Ctrl + Alt + Del.
    - Step 2: Choose Restart.

    - Step 3: If issue persists, boot into Safe Mode.
- **Peripheral Issues**
    - Step 1: Check cable connections.
    - Step 2: Open Device Manager → Update driver.
    - Step 3: Test device again.
- **Error Messages**
    - Step 1: Note error code.
    - Step 2: Search official documentation or support forums.

5. Practical Exercises (Day 4)

- Perform proper startup and shutdown.
- Put system into Sleep, resume, check apps.
- Repeat with Hibernate, compare differences.
- Create a new account, log in, and test permissions.
- Simulate frozen app → close via Task Manager.
- Disconnect printer → reconnect and reinstall driver.